

Cover Letter: Nature Methods Submission

To: The Editors, Nature Methods

From: Gary Welz

Date: January 2026

Subject: Submission of Method Article: “The Programming Framework: A General Method for Process Analysis Using LLMs and Mermaid Visualization”

Dear Editors,

I am submitting a Method article entitled “The Programming Framework: A General Method for Process Analysis Using LLMs and Mermaid Visualization” for consideration for publication in Nature Methods.

Why this paper fits Nature Methods:

This paper presents a methodological contribution that addresses a fundamental challenge in scientific communication and knowledge representation: transforming textual process descriptions into structured, computable visualizations that work across multiple scientific disciplines. Nature Methods is the ideal venue for this work because it publishes methodological advances with broad applicability, and the Programming Framework demonstrates precisely that—a general methodology successfully applied across biology (100+ processes), chemistry (70+ processes), mathematics, physics, and computer science.

Methodological Innovation:

The Programming Framework provides a systematic, LLM-based methodology for extracting and visualizing scientific processes that has been successfully applied across five distinct scientific domains. While individual components (large language models, Mermaid diagramming, process visualization) are well-established, our contribution is the synthesis of these elements into a coherent, repeatable methodology with demonstrated cross-domain applicability. The framework’s suggested five-category color-coding system provides a starting point for consistent representation while allowing domain-specific customization, as demonstrated in the Genome Logic Modeling Project’s specialized biological regulatory networks.

Broad Applicability and Impact:

We demonstrate the framework’s effectiveness through comprehensive application to: - **Biology:** 100+ processes via the Genome Logic Modeling Project, covering central dogma, metabolism, signaling, protein processes, photosynthesis, and DNA repair - **Chemistry:** 70+ processes across organic reactions, synthesis pathways, thermodynamics, catalysis, and electrochemistry - **Mathematics:** Algorithms including the Euclidean algorithm and proof methods - **Physics:** Quantum mechanics, nuclear physics, electromagnetism, and thermodynamics -

Computer Science: Algorithms, software engineering, AI, security, networks, databases, and graphics

This breadth of application demonstrates that the methodology is not domain-specific but provides a general approach to process visualization and analysis that researchers across scientific disciplines can adopt.

Reproducibility and Open Availability:

All methodology, tools, and examples are publicly available: - Framework implementation and documentation: https://huggingface.co/spaces/garywelz/programming_framework
- Biological applications (GLMP): <https://huggingface.co/spaces/garywelz/glmp>
- All figure source code (Mermaid Markdown) and processed data (JSON) are accessible through these repositories - Complete prompt templates are provided in the manuscript appendix

The methodology is fully documented with prompt templates, validation workflows, iterative refinement procedures, and reproducible examples, supporting Nature Methods' commitment to reproducible research and methodological transparency.

Practical Impact:

The Programming Framework addresses a critical gap in scientific process representation: while domain-specific standards exist (SBGN for biology, UML for software), there is no accessible, cross-disciplinary approach that enables researchers to visualize and compare processes across domains. This framework provides that infrastructure, with demonstrated applications in research, education, and AI system integration. The methodology is designed as reusable infrastructure that others can adopt, extend, and critique, rather than a closed system.

Significance for the Methods Community:

This methodological contribution will be valuable to researchers who: - Need to visualize and compare processes across disciplines - Work with textual process descriptions that require structured representation - Develop educational materials requiring clear process visualizations - Build AI systems that need to understand and represent scientific processes - Seek reproducible methods for knowledge representation

The framework's text-based format (Mermaid Markdown) enables version control, collaborative editing, and seamless integration into documentation systems, making it accessible to researchers without specialized software or training.

Article Type:

This is submitted as a **Method** article, presenting a novel methodological approach with comprehensive empirical demonstrations across multiple scientific domains.

Preprint Status:

This manuscript has not been posted as a preprint. We plan to post it to a preprint server after journal submission if permitted.

Author Information:

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Data Availability:

All data and materials are publicly available at the repositories listed above. The manuscript includes complete methodology documentation, prompt templates in the appendix, validation procedures, and links to all supporting materials including the live, interactive web applications demonstrating the framework.

Code Availability:

Source code and methodology implementation are publicly available at: https://huggingface.co/spaces/garywelz/programming_framework. Additional code and examples are available at: <https://huggingface.co/spaces/garywelz/glmp>. All prompt templates and validation workflows are documented in the manuscript appendix.

Related Manuscripts:

No related manuscripts are under consideration elsewhere.

Thank you for considering this submission. I believe this methodological contribution will be of significant interest to the Nature Methods readership, particularly researchers working across disciplines who need accessible, reproducible methods for process visualization and analysis. I look forward to your feedback and the opportunity to contribute to Nature Methods' mission of publishing methodological advances with broad applicability.

Sincerely,
Gary Welz

Manuscript: - Main manuscript: Single PDF file with all figures embedded as recommended by Nature Methods submission guidelines - The manuscript includes 4 figures embedded within the text: - Figure 1: Basic Programming Framework Structure - Figure 2: Beta-Galactosidase Regulation System - Figure 3: Catalytic Hydrogenation Reaction - Figure 4: Euclidean Algorithm